

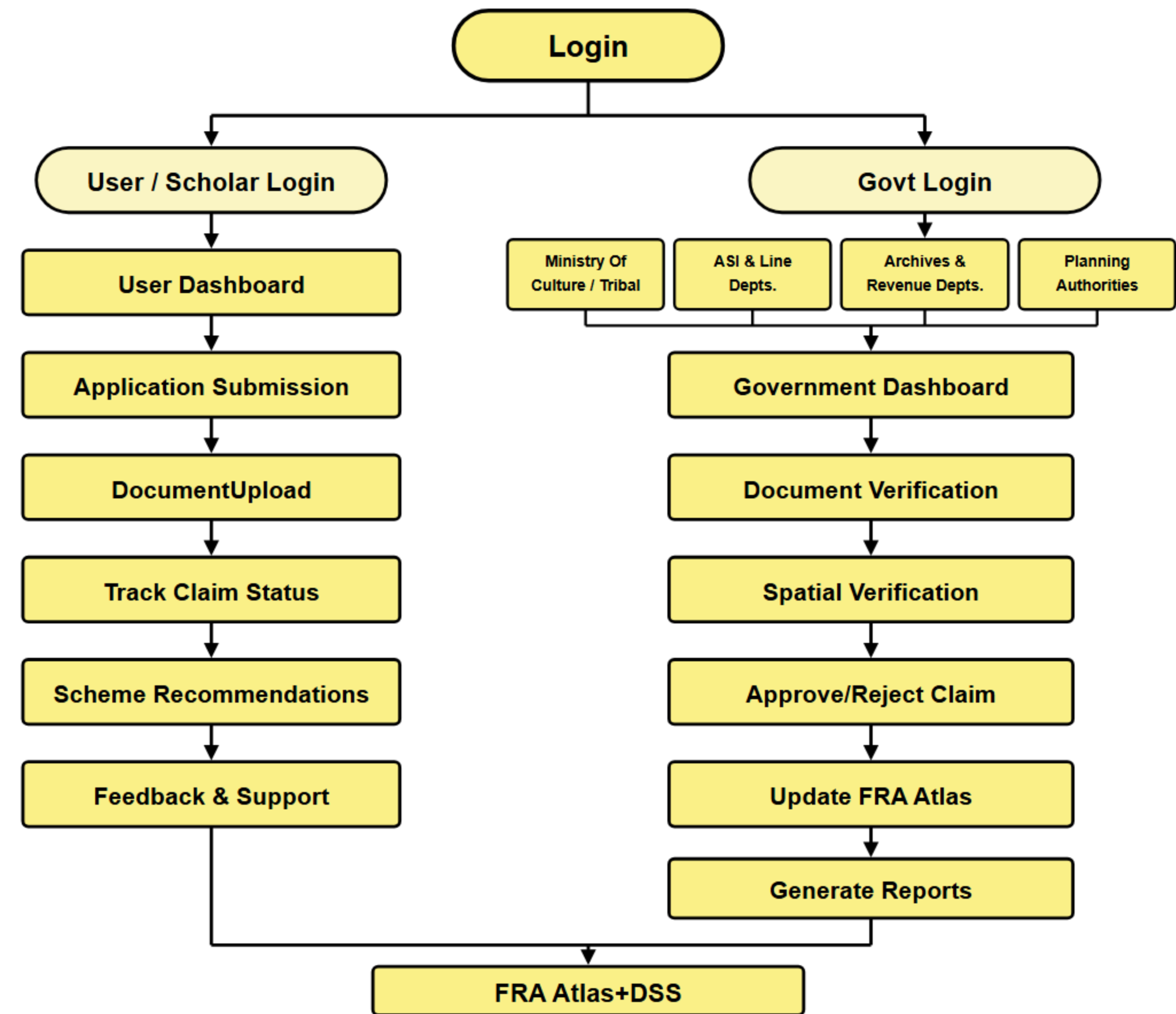
TITLE PAGE

- **Problem Statement ID** – SIH26195
- **Problem Statement Title**- Digital Preservation, AI-Powered Paleography and Interactive 3D WebGIS Atlas for Indian Cultural Heritage, Monuments & Ancient Manuscripts. (Centres to be concentrated: National Archives, ASI Sites, Craft Clusters)
- **Theme**- Heritage & Culture
- **PS Category**- Software
- **Team ID**- 57385
- **Team Name**- Sanskriti RakshaX



❖ Proposed Solution:

- **3D Archival & Artifact Studio :**
360° interactive rotation, dynamic lighting simulation & sub-millimeter loupe inspection for sculptures.
- **Ancient Manuscript Paleography :**
AI transcription of fragile palm-leaf (Talapatra) & birch-bark (Bhojpatra) scripts with Sanskrit chant audio.
- **Multimodal AI Heritage Guide :**
Voice, vision, and multilingual conversational assistance grounded in verified ASI/museum archives.
- **Computer Vision Iconography Scanner :**
Identifies dynastic art styles, postures (asanas), mudras, and motifs from user photos.
- **Chronological Epoch Explorer :**
Interactive 11-epoch historical timeline spanning 5,000+ years of civilizational heritage.
- **Interactive Heritage Web-GIS Atlas :**
Dynamic geospatial platform for exploring monuments, excavations, and conservation zones.
- **GI Artisan Marketplace & Livelihood :**
Direct fair-trade connect supporting GI-certified traditional master artisan cooperatives.
- **Sustainable Tourism & Itinerary AI :**
Eco-friendly tour planner dispersing crowds away from overtouristed monuments.
- **GI Tagging and E-Signature :**
To preserve cultural heritage, safeguard traditional handicraft products and paperless verification.



- **Data Extraction** : Optical Character Recognition(OCR), NLP.
- **WebGIS platform** : Leaflet.js(for interactive maps).
- **Asset Mapping** : Random Forest & CNN.
- **Data Storage** : PostgreSQL & PostGIS(for geospatial data).
- **Data Management** : Cloud Object Storage & MySQL(Structured).
- **GeoServer** : Use standard protocols WMS and WFS to serve maps.
- **Decision Support System** : for scheme recommendations.
- **Blockchain based Security** : Hyperledger Fabric.
- **Key Lock Framework** : Open-source Identity & Access Management
- **Rasa** : Preferred for its offline, multilingual, and secure chatbot.



FRA Setu / VirasatX Portal (Working Prototype)

English / हिन्दी / தமிழ்

Nav

- Home / Login
- Claims & Maps
- Downloads
- Help & Support

Interactive FRA Map / 3D Canvas

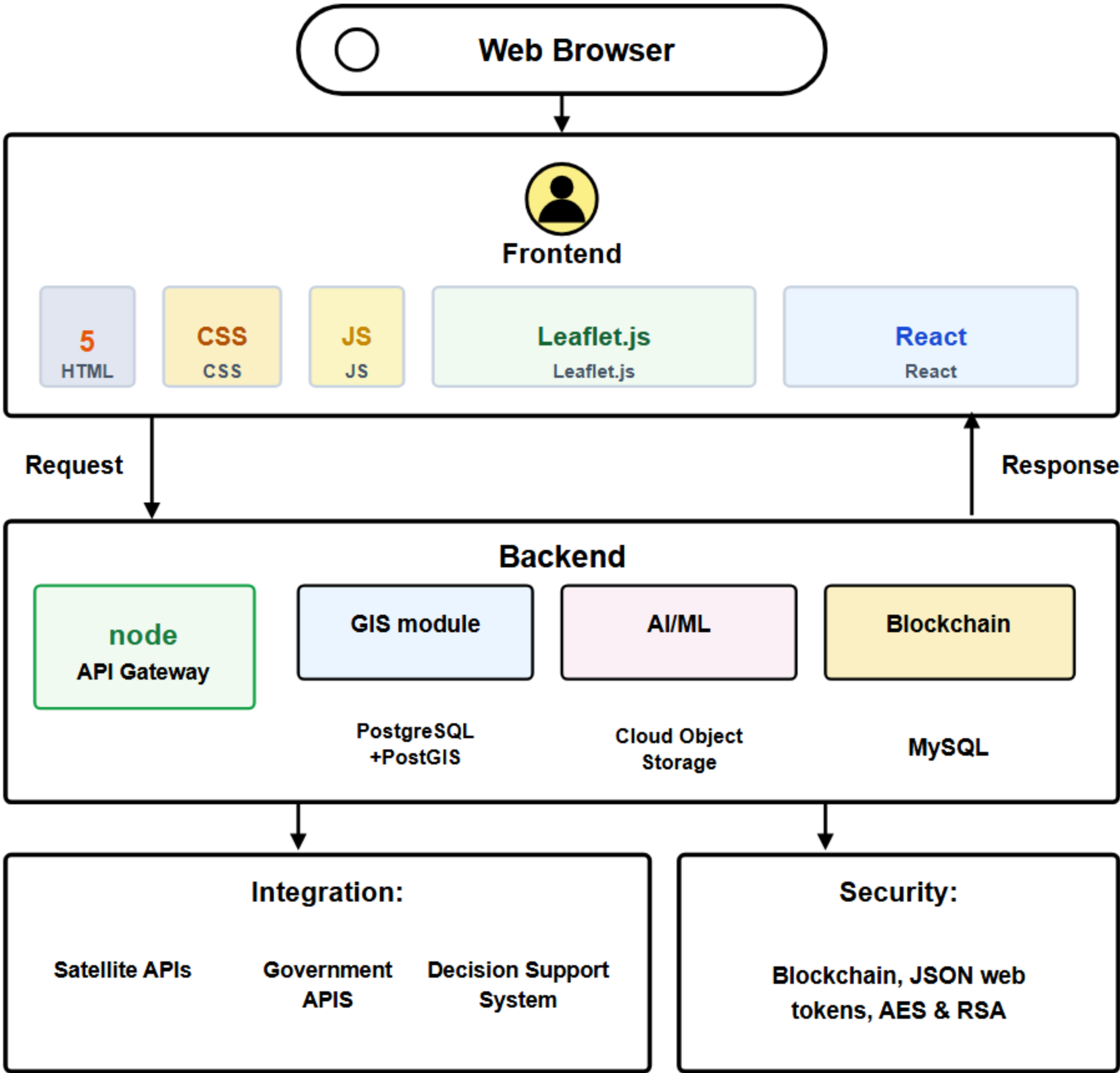
[Real-time Layer Rendering | GeoServer]

1250 Verified 89 Rejected

Grievance / DSS

- AI Scheme Match
- Officer Tracker
- 2.1 Avg Days

User Interface: It is a draft version, subject to future updates.



❖ Feasibility:

1. Technical Feasibility:

1. Digitization, AI Mapping and Web-GIS Dashboards:

- OCR and AI extract data from fragile ancient palm-leaf & birch-bark manuscripts.
- Web-GIS dashboards show real-time interactive heritage maps and 3D models.

2. DSS Integration with Blockchain Security:

- AI engines match cultural schemes and tourist traffic to community needs.
- Blockchain ensures secure, tamper-proof digital records and artisan GI credentials.

2. Operational Feasibility:

1. Existing Setup and Simple Staff Training:

- Museum & archival data exists, needs unified digital integration.
- Staff need basic training through short interactive workshops.

2. Stakeholder Support and Effective Implementation:

- Ministries, ASI, and cultural NGOs already support heritage preservation.
- System strengthens existing efforts, seamlessly augmenting current curation work.

3. Legal & Ethical Feasibility:

1. Data Privacy & IP Safeguards:

- Must follow India's DPDP Act, ensuring legal compliance and encryption.
- Blockchain and watermarking safeguard sensitive indigenous IP and artisan rights.

2. Consent Mechanism:

- Informed consent and transparent citations required in all archival processes.
- Real-time feedback enables secure, two-way curatorial communication.

❖ Viability:

1. Economic Viability:

1. Cost-Effective & Efficient:

- Saves paperwork, manual digitizing effort, duplication, and reduces physical damage.
- Cuts curation delays, transport expenses, and institutional overhead.

2. Supported & Scalable Technology:

- Backed by national digital missions and cultural tourism initiatives (SDG 8 & 11).
- Open-source tech ensures flexible, low-cost, future-ready deployment.

2. Social Viability:

1. Empowers Communities & Improves Heritage Access:

- Provides transparency, secure records, and interactive learning for 250M+ youth.
- Directly benefits traditional GI master artisans through fair-trade market access.

2. Reduces Conflict & Promotes Stronger Inclusion:

- Verified records and academic citations reduce disputes in historical interpretations.
- Mobile updates, voice guides, and regional languages improve grassroots engagement.

3. Scalability:

1. Pan-India Expansion for Wider Community Benefits:

- Pilot in key museums & ASI circles, then scale seamlessly across India.
- Adjust platform easily for state-level regional languages and local folklore.

2. Cross-Sector Use of WebGIS + DSS:

- Adaptable for archaeological survey, urban heritage planning, and education.
- Useful in disaster management, structural monitoring, and post-crisis restoration.

❖ Benefits:

1. Social Benefits:

- Digital platform empowers 250M+ students with interactive 3D civilizational exploration.
- IVR helpline, voice guide, and 8 regional languages ensure inclusive literacy access.
- Transparent records, authentic audio chants, and museum dossiers build cultural pride.

2. Environmental Benefits:

- Sustainable cultural tourism with AI-guided crowd dispersal away from fragile monuments.
- Digital twins preserve physical integrity of deteriorating stone reliefs and manuscripts.
- GI Tagging preserves cultural heritage, eco-friendly crafts & traditional products.

3. Economic Benefits:

- Access to schemes (National Heritage City Development, PRASHAD, Swadesh Darshan).
- GI Tagging boosts local artisanal products & tribal livelihoods (SDG 8).
- Direct artisan connect ensures economic stability -> higher wages & eliminating middlemen.

4. Governance Benefits:

- Audit trails ensure curatorial accountability, prevent asset theft or neglect.
- Curatorial review portal boosts transparency, prevents illicit antiquities trade.
- Cross-validation stops duplicate records, misinformation, and ungrounded AI claims.

5. Legal & Security Benefits:

- E-Signature & Digital Records provide legally valid archival accession provenance.
- Protection against illicit duplication and intellectual property theft.
- Digital witness/testimony ensures fair scholarly consensus and dispute settlement.

❖ Impacts:

1. Secure Cultural Rights & Transparency:

- Communities and researchers track heritage data, reducing fraud and neglect.
- Digital catalog ensures authenticity in national and global archival records.
- Builds institutional trust between citizens, scholars, and government authorities.

2. Heritage Protection & Conservation with AI:

- AI monitors structural and digital assets to prevent loss and decay.
- Conserves civilizational knowledge and safeguards ancient Indian manuscripts.
- Promotes sustainable, community-driven cultural resource management.

3. Boost Livelihoods & Local Development:

- Resource planning improves artisan incomes and rural handicraft clusters.
- GI tagging protects traditional knowledge, boosts global market branding.
- VirasatX enables fair access to cultural preservation grants and welfare schemes.

4. Digital, Eco-Friendly & Cost Effective:

- Reduces paper usage, ensuring greener, eco-friendly e-governance (SDG 11).
- Preserves indigenous crafts through high-fidelity digital archiving tools.
- Lowers administrative and physical curation costs while maximizing outreach.

5. Smart, Transparent & Scalable Governance:

- Transparent dashboards track museum digitizing progress and scholar engagement.
- AI dispute mediation ensures historical fairness, reducing subjective human bias.
- Unified digital platform enables future-ready, pan-India expansion (Viksit Bharat 2047).

1. Archaeological Survey of India (ASI) & National Heritage Portals:

The Archaeological Survey of India (ASI) and Ministry of Culture govern the protection and documentation of cultural heritage monuments, ancient inscriptions, and archaeological excavations.
<https://asi.nic.in/> | <https://asi.nic.in/epigraphy/>

2. National Mission for Manuscripts (NMM) & Paleography Standards:

Established by Ministry of Culture to unearth, preserve, and digitally catalog India's vast manuscript heritage (Talapatra, Bhojpatra) and ancient epigraphy.
<https://www.namami.gov.in/> | <https://ignca.gov.in/>

3. Best Practices for Web Security & Cryptographic Integrity:

This framework outlines key web security measures, DPDP Act 2023 compliance, and verifiable tamper-proof records using Blockchain & Encryption Standards.
<https://owasp.org/www-project-top-ten/> | <https://www.meity.gov.in/data-protection-framework>

4. The AMASR Act 1958 & GI Registry:

The Ancient Monuments and Archaeological Sites and Remains Act (1958 & 2010) and Geographical Indications of Goods Act (1999) safeguarding cultural heritage & traditional artisan products.
<https://indiacode.nic.in/handle/123456789/1631> | <https://ipindia.gov.in/geographical-indications.htm>

To enable smooth implementation and nationwide integration of the VirasatX Digital Heritage System, the following contacts and portals serve as official national-level resources for communication, data coordination, and institutional support:

5. Ministry of Culture: <https://indiaculture.gov.in/>

7. National Archives of India: <https://www.abhilekh-patal.in/>

6. National Museum, New Delhi: <https://nationalmuseumindia.gov.in/>

8. UNESCO World Heritage Centre: <https://whc.unesco.org/en/statesparties/in>